HH–P–131c
SEPTEMBER 13, 1954

SUPERSEDING
Int. Fed. Spec. HH–P–131c (Navy–Ships)
19 October 1951, and
Fed. Spec. HH–P–131b
December 18, 1947

FEDERAL SPECIFICATION

PACKING; METALLIC AND NONMETALLIC, PLASTIC

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers packings for use under the following conditions:

Rods and valve stems:

Steam to 300 pounds pressure and 260° C. (500° F.) temperature: Types I and II, classes 1 and 2.

Steam to 300 pounds pressure and temperatures over 260° C. (500° F.): Type III, class 2.

Water and oil to 500 pounds pressure and 260° C. (500° F.) temperature: Types 1 and II, classes 1, 2, and 3.

Brine to 100 pounds pressure: Types I and II, classes 1, 2, and 3.

Air to 200 pounds pressure: Types I and II, classes 1, 2, 3, and 4.

Gas to 150 pounds pressure: Types I and II, classes 1, 2, and 3.

Expansion and slip joints:

Steam to 300 pounds pressure and 370° C. (700° F.) temperature: Types II and III, classes 2 and 4.

1.2 CLASSIFICATION

1.2.1 *Types and classes.*—Plastic, metallic and nonmetallic packing covered by this specification shall be of the following types and classes, as specified in the invitation for bids:

Type I.—Plastic, metallic, lead base.
Class 1.—Bulk form.
Class 2.—Coil form, with or without cotton jacket.
Class 3.—Coil form, metal-foil jacket.
Class 4.—Coil form, braided-wire jacket.

Type II.—Plastic, nonmetallic.
Class 2.—Coil form, cotton jacket.
Class 3.—Coil form, braided-wire-and-asbestos jacket.
Class 4.—Coil form, braided-wire jacket.

Type III.—Plastic, metallic, copper base.
Class 2.—Coil form, cotton jacket.

1.2.2 *Sizes.*—Plastic, metallic and nonmetallic packing covered by this specification shall be furnished in the sizes specified in the invitation for bids.

2. APPLICABLE SPECIFICATIONS

2.1 There are no other specifications applicable to this specification, except those referenced in section 7 by individual departments.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

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3. REQUIREMENTS

3.1 Material.—The metallic packing shall consist, in general, of metal particles, asbestos fiber, graphite, and binder. The composition of the nonmetallic packing shall consist, in general, of asbestos fiber, graphite, and binder.

3.2 Binder.—The different components of class 1 and of the core of classes 2, 3, and 4 shall be thoroughly mixed with a binder of a material suitable for the temperature conditions for which the packing is required; and shall be so held together by the binder as to prevent separation of the various parts.

3.3 Graphite.—Graphite, when used for lubrication, shall contain not less than 90 percent, by weight, of graphite carbon.

3.4 Flexibility.—The flexibility of the packing shall be such as to permit it to be readily formed into coils to fit a rod of the diameter shown in table I.

TABLE I.—*Size of coils*

Size of packing	Rod diameter	Size of packing	Rod diameter
Inch	*Inches*	*Inches*	*Inches*
Up to ½, inclusive	3	1	8
⅝	4	1⅛	9
¾	5	1¼	10
⅞	6		

3.5 Type I, plastic, metallic, lead base.—The metal used in lead-base packing shall not melt or disintegrate at a temperature below 288° C. (550° F.) when tested as described in 4.3.4. It shall be in one or more of the following forms: finely ground, small pellets, small strands, or small shreds.

3.5.1 *Class 1.*—Class 1 packing shall be furnished in bulk form.

3.5.2 *Class 2.*—Class 2 packing shall be furnished in continuous strips of approximately uniform cross section, partially compressed to prevent disintegration during application. It may be enclosed in a latticed cotton thread or knit cotton jacket or may be furnished without a covering.

3.5.3 *Class 3.*—Class 3 packing shall be furnished in continuous strips of uniform cross section, completely enclosed in a metal-foil jacket consisting of not less than two wraps on sizes up to ⅜ inch and three wraps on larger sizes. The metal foil shall be not less than 0.0044 inch thick.

3.5.4 *Class 4.*—Class 4 packing shall be furnished in continuous strips of uniform cross section, extruded into a braided-wire jacket. The jacket shall be of copper wire not more than 0.008 inch in diameter.

3.6 Type II, plastic, nonmetallic.

3.6.1 *Class 2.*—Class 2 packing shall be furnished in continuous strips of approximately uniform cross section, partially compressed to prevent disintegration during application, and enclosed in a latticed cotton thread or knit cotton jacket.

3.6.2 *Class 3.*—Class 3 packing shall be furnished in continuous strips of uniform cross section, completely enclosed in a braided jacket composed of asbestos fiber and nickel-copper wire. The asbestos fiber shall contain not less than 12 per cent, by weight, of chemically combined water.

3.6.3 *Class 4.*—Class 4 packing shall be furnished in continuous strips of uniform cross section, extruded into a braided-wire jacket. The jacket shall be of copper wire not more than 0.008 inch in diameter.

3.7 Type III, plastic, metallic, copper base.—The metal shall not melt or disintegrate at a temperature below 650° C. (1200° F.) when tested as described in 4.3.4. It shall be in one or more of the following forms: finely ground, small pellets, small strands, or small shreds.

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3.7.1 *Class 2.*—Class 2 packing shall be furnished in continuous strips, of approximately uniform cross section, partially compressed to prevent disintegration during application, and enclosed in a latticed cotton thread or knit cotton jacket.

3.8 Workmanship.—Workmanship shall be first class in every respect.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.—A 1-pound sample of class 1 and a sample of sufficient length of classes 2, 3, or 4 to make the coil test, shall be taken at random from each class 1 and from each size delivered for inspection and test. An additional piece 4 inches long shall be taken from each size for determining chemically combined water and graphite content.

4.2 Inspection.—Packing shall be inspected to determine compliance with this specification with respect to material, workmanship, construction, packaging, packing, and marking.

4.3 Tests.

4.3.1 *Chemically combined water.*—A specimen weighing approximately 1 gram shall be taken from the fibrous material in the jacket of type II, class 3 packing. It shall be placed in a porcelain or platinum combustion boat, dried for 1 hour at a temperature of 105° to 110° C., cooled in a desiccator, and weighed. The dried specimen in the boat shall be inserted in the combustion tube of an electric, organic combustion furnace. The specimen shall be maintained at a temperature of 900° ± 50°C. for approximately 30 minutes or until combustion of the cotton is complete. During the combustion period a current of oxygen (carbon-dioxide free) shall be passed through the combustion tube at a rate of approximately 200 milliliters per minute. The combustion gases shall be passed through either two U-tubes containing calcium chloride or through a drying tube containing anhydrous magnesium perchlorate or calcium sulphate to remove the moisture; and finally the gases shall be passed into either a weighed Vanier or similar absorption bulb containing a strong solution of caustic potash, or in a weighed carbon-dioxide absorption bulb containing a sodium-hydroxide impregnated base (the absorbent having the trade name "Ascarite" is of this type) to absorb the carbon dioxide. Three-elevenths of the increase in weight of the Vanier or other carbon-dioxide absorption bulb shall represent the weight of the carbon in the cloth or yarn. This shall be 44.40 per cent of the cotton. These factors may be combined to give a constant of 0.614. When the combustion has been completed, the absorption tube shall be weighed, and the combustion boat containing the ignited asbestos shall be removed from the furnace, cooled in a desiccator, and weighed.

4.3.1.1 *Calculation.*—

$$\text{Cotton, percent, } A = \frac{61.4 \times C}{S}$$

$$\text{Ignited asbestos, percent, } B = \frac{R}{S} \times 100$$

Chemically combined water in the asbestos, percent

$$= \frac{100 - A - B}{100 - A} \times 100$$

where S = weight of specimen of fibers.
C = weight of carbon dioxide absorbed, 4.3.1.
R = weight of ignited asbestos, 4.3.1.
A = percent cotton.
B = percent ignited asbestos.

4.3.1.2 Two specimens shall be tested. The average of the results obtained from the two specimens shall be the asbestos content of the sample.

4.3.2 *Graphitic carbon, by combustion method.*—A specimen weighing from 0.200 to 0.300 gram shall be placed in a small Erlenmeyer flask, about 25 milliliters of ethyl ether added, corked loosely, and allowed to stand for about ½ hour, shaking at intervals. The mixture shall be filtered onto asbestos in a Gooch crucible, the asbestos having been previously washed with hydrochloric acid and ignited. The residue on the filter shall be washed with alcohol and next with distilled water. The residue shall then be washed with hot (1.1) hydrochloric acid

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to remove any carbonates that might be present and then with hot distilled water until free of chlorides. The filter and the residue shall then be transferred to a porcelain or platinum boat and dried on the hot plate. The boat containing the residue shall be transferred to the tube of a combustion furnace and burned at a temperature of 800° to 850° C. (1470° to 1560° F.) in a stream of oxygen gas. The carbon dioxide which forms shall be collected in a potash bulb with a 30 percent potassium-hydroxide solution. The combustion tube shall contain some fused lead chromate to retain any sulphur that may be present. If in doubt as to whether the carbon is graphite, graphite carbon may be identified by the use of the electron microscope or from X-ray diffraction patterns.

4.3.3 *Coil test.*—The coil test for flexibility shall be made by hand. Packing shall be coiled around a rod of the diameter shown in table I. Only sufficient packing to make two complete turns around the rod, and 4 inches at each end for hand grips, shall be used.

4.3.4 *Heat test (for types I and III only).*—A small specimen shall be cut from each sample and subjected for 2 hours to a temperature of 288° C. (550° F.) for type I packing and 650° C. (1200° F.) for type III packing, as applicable.

5. PREPARATION FOR DELIVERY

5.1 Packaging.—Unless otherwise specified, commercial packages are acceptable under this specification.

5.2 Packing.—Unless otherwise specified, the subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to the point of delivery called for in the contract or purchase order. The gross weight of a single container shall not be in excess of approximately 300 pounds. Only one type, class, and size shall be packed in a single container.

5.3 Marking.

5.3.1 *Packages.*—Unless otherwise specified, packages shall be marked with the name of the material, the type, class, size, and quantity contained therein, as defined by the contract or order under which shipment is made, and the name of the manufacturer.

5.3.2 *Shipping containers.*—Unless otherwise specified, shipping containers shall be marked with the name of the material, the type, class, size, and quantity contained therein, as defined by the contract or order under which shipment is made, the name of the contractor, the number of the contract or order, and the gross weight.

6. NOTES

6.1 Purchasers should exercise any desired options offered herein and should specify the type, class, size, and the quantity in pounds required. The usual commercial practice is to sell by the pound. (See 1.2.1, 1.2.2, 5.1, 5.2, 5.3, and sec.7.)

6.2 Transportation description.—The proper transportation description

Type I, class 1:
Packing, metal.
Carload weight 30,000 pounds.
Truckload weight 30,000 pounds.

Type I, classes 2, 3, 4; type III, class 2:
Packing forms or shapes, not otherwise indexed by name.
Carload weight 30,000 pounds.
Truckload weight 24,000 pounds.

Type II, classes 2, 3, 4:
Packing forms or shapes, not otherwise indexed by name, other than metal or rubber.
Carload weight 30,000 pounds.
Truckload weight 30,000 pounds.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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7. DEPARTMENTAL REQUIREMENTS

7.1 The following specifications and standard, of the issues in effect on date of invitation for bids, and special requirements, form a part of this specification for purchases made under this specification by the respective departments.

7.2 Army and Air Force.

7.2.1 *Applicable specifications and standard.*

Federal Specifications:

NN–B–591—Boxes; Fiberboard, Wood-Cleated (for Domestic Shipment).

NN–B–621—Boxes; Wood, Nailed and Lock-Corner.

NN–B–631—Boxes; Wood, Wirebound (for Domestic Shipment).

LLL–B–631—Boxes; Fiber, Corrugated (for Domestic Shipment).

LLL–B–636—Boxes; Fiber, Solid (for Domestic Shipment).

PPP–B–601—Boxes, Wood, Cleated-Plywood.

Military Specifications:

JAN–P–105—Packaging and Packing for Overseas Shipment—Boxes, Wood, Cleated, Plywood.

JAN–P–106—Packaging and Packing for Overseas Shipment—Boxes; Wood, Nailed.

MIL–B–107—Boxes, Wood, Wirebound (Overseas Type).

JAN–P–108—Packaging and Packing for Overseas Shipment—Boxes, Fiberboard (V-Board and W-Board), Exterior and Interior.

JAN–P–125—Packaging and Packing for Overseas Shipment—Barrier-Materials, Waterproof, Flexible.

JAN–P–138—Packaging and Packing for Overseas Shipment—Boxes, Wood, Fiberboard-Lined.

MIL–A–140—Adhesive, Water-Resistant, Waterproof Barrier-Material.

JAN–P–139—Packaging and Packing for Overseas Shipment—Plywood, Container-Grade.

MIL–P–17303—Packing Material, Plastic, Metallic and Nonmetallic.[1]

Military Standard:

MIL–STD–129—Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

7.2.2 *Packaging.*—Unless otherwise specified, commercial packages are acceptable under this specification.

7.2.3 *Packing.*

7.2.3.1 *For domestic shipment.*—The subject commodity packaged as specified in 7.2.2 shall be packed in wood-cleated fiberboard, wood-cleated plywood, nailed wood, wirebound wood, corrugated, or solid fiberboard boxes conforming to Federal Specification NN–B–591, PPP–B–601, NN–B–621, NN–B–631, LLL–B–631, or LLL–B–636, respectively. The gross weight of wood boxes shall not exceed approximately 200 pounds; of fiberboard boxes, approximately 70 pounds.

7.2.3.2 *For overseas shipment.*—The subject commodity packaged as specified in 7.2.2 shall be packed in cleated-plywood boxes, nailed wood boxes, wirebound wood boxes, fiberboard boxes, or fiberboard lined wood boxes conforming to Military Specifications JAN–P–105, style A or B, JAN–P–106, style 2, 2½ or 3, MIL–B–107, JAN–P–108, symbol V3c or V3s, and JAN–P–138, respectively. When plywood boxes are used, the plywood shall conform to type A or B, condition 1 of Military Specification JAN–P–139. Shipping containers shall be lined with a sealed waterproof bag made from material conforming to Military Specification JAN–P–125. The seams and closures shall be sealed with water-resistant adhesive conforming to Military Specification MIL–A–140. The gross weight of wood boxes shall not exceed approximately 200 pounds; of fiberboard boxes, approximately 70 pounds.

[1] Applicable to Navy purchases only.

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7.2.4 *Marking.*—In addition to any special marking required by the contract or order, marking of interior packages and shipping containers shall be in accordance with Military Standard MIL–STD–129.

7.3 Navy purchases.—Navy purchases shall be made under Military Specification MIL–P–17303, Packing Material, Plastic, Metallic and Nonmetallic, which entails the use of a qualified products list.

Copies of this specification may be purchased for ents

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